

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA
National Mathematics Talent Contests – Screening Test
Mathematics Olympiad at Sub-Junior Level for VII & VII Classes
Kaprekar Contest – Saturday, August 30, 2003

Note: Read carefully the instructions given in the Response Sheet.
 Duration of the test 2pm to 4p.m

01) In the given multiplication, a and b are natural numbers.

Then $a + b$ is

A) 5

B) 6

C) 8

D) 9

$$\begin{array}{r} 3a \\ \times b2 \\ \hline 70 \\ 140 \\ \hline 1470 \end{array}$$

02) The largest value among $\frac{4}{5}$, 80%, 0.801 and $(.9)^2$ is

A) $\frac{4}{5}$

B) 80%

C) 0.801

D) $(.9)^2$

03) If the sum of two consecutive odd numbers is 2004, then the smaller of the two numbers could be

A) 2001

B) 1001

C) 1003

D) 1

04) If x is $\frac{5}{7}$ of 490, then $\frac{5}{7}$ of x is

A) 350

B) 250

C) 70

D) 420

05) If $x^{-1} = 4^{-1} + 5^{-1}$, then x is

A) $\frac{20}{9}$

B) $\frac{1}{9}$

C) $\frac{9}{20}$

D) $\frac{-1}{9}$

06) If $2a - 3 = 5$ and $3b + 1 = 2$, then $3b - 2a$ is

A) - 3

B) - 7

C) 7

D) - 9

07) Rahim wants to arrange a party of a certain number of people such that no two of the participants (whose dates of birth he knows) of the party will have birthdays in the same month. The number of people to be invited for the party cannot exceed

A) 12

B) 13

C) 364

D) 23

08) If $x = 9 + 4\sqrt{5}$ and $xy = 1$ then $\frac{1}{x^2} + \frac{1}{y^2}$ is

- A) 81 B) 322 C) 97 D) 2

09) In a kilometre race, A beats B by 1 minute and beats C by 375 metres. If B beats C by 30 seconds, the time taken by C to run 1 km is

- A) 150 sec. B) 210 sec. C) 240 sec. D) 200 sec.

10) Turning a screw driver by 90° , we can drive a screw 5 mm deeper into a piece of wood. How many complete revolutions are needed to drive the screw 6 cm into wood?

- A) 2 B) 3 C) 4 D) 12

11) The sum $1 - 2 + 3 - 4 + \dots + 47 - 48 + 49$ is

- A) 0 B) 24 C) 25 D) $\frac{49 \times 50}{2}$

12) If $4^{44} + 4^{44} + 4^{44} + 4^{44} = 4^x$ then x is

- A) 45 B) 44 C) 176 D) 11

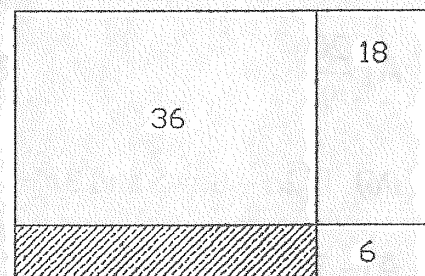
13) The *units digit* of the sum of the first 99 whole numbers (that is, $0+1+2+\dots+98$) is

- A) 0 B) 1 C) 5 D) 9

14) Which one of the following is a perfect square?

- A) $4^5 5^4 6^6$ B) $4^6 5^4 6^5$ C) $4^4 5^5 6^6$ D) $4^4 5^6 6^5$

15) A rectangle is divided into four smaller rectangles, as shown in the figure. Three of them have areas 6, 18 and 36, as indicated. The area of the shaded rectangle is



- A) 30 B) 24
C) 12 D) 18

16) Today is Saturday. What day of the week is 100 days from now?

- A) Friday B) Tuesday C) Monday D) Sunday

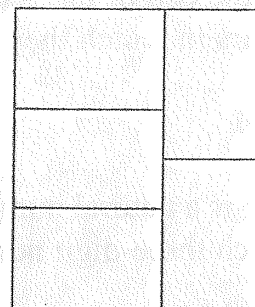
17) A brick weighs $\frac{3}{4}$ th the weight of itself plus $\frac{3}{4}$ kg. The weight of the brick (in kg) is

- A) 2 B) 2.5 C) 3 D) 2.75

18) The cost of a single two-litre bottle of juice is 5 times the cost of 1 cup of juice. If a single two-litre bottle of juice costs Rs.6 more than 3 cups of juice, how much does a single two-litre bottle of juice cost?

- A) Rs.10 B) Rs.12 C) Rs.15 D) Rs.18

19) A rectangle with perimeter 44 is partitioned into 5 congruent rectangles, as indicated in the diagram. The perimeter of each of the congruent rectangles is



- A) 10 B) 20
C) 24 D) 48

20) The sum of seven consecutive natural numbers is 84. The difference between the largest and the smallest among them is

- A) 7 B) 6 C) 14 D) 21

21) If each diagonal of a square has length 2, then the area of the square is

- A) 2 B) $4\sqrt{2}$ C) 1 D) $2\sqrt{2}$

22) Four points P, Q, R, S are on a line segment, as shown.



If $PQ : QR = 1 : 2$, $QR : RS = 8 : 5$, then $PQ : QS$ is

- A) 3:13 B) 1:7 C) 1:13 D) 4:13

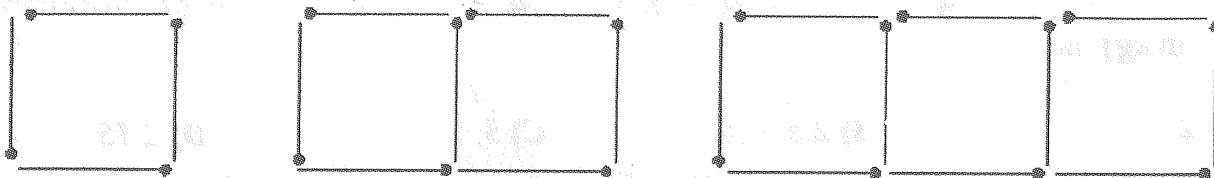
23) If $a+b+c=0$, then $\frac{b^2+c^2+a^2}{b^2-ca}$ is

- A) 0 B) 1 C) 2 D) $3b$

24) $ABCD$ is a square. P is a point in its interior such that $\triangle PAB$ is equilateral. Then measure of $\angle PCB$ is

- A) 45° B) 30° C) 60° D) 75°

25) Match-sticks are used to form the following pattern made of squares:



In this pattern, how many match-sticks are needed to form ten squares in a row?

- A) 36 B) 40 C) 31 D) 30

26) At the end of the year 2002, Ram was half old as his grandpa. The sum of the years in which they were born is 3854. Age of Ram at the end of year 2003 is

- A) 50 B) 35 C) 51 D) 36

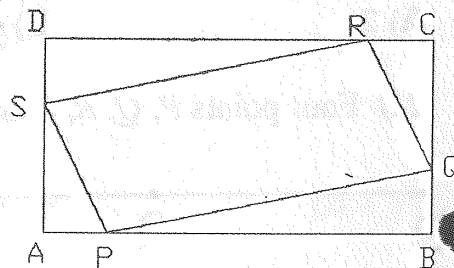
27) 146 is a natural number whose “digit-product” is $1 \times 4 \times 6 = 24$. How many such three-digit numbers are possible with “digit-product” 24 ?

- A) 4 B) 18 C) 21 D) 16

28) I multiplied a natural number by 18 and another natural number by 21. Then I added the products. Which one of the following could be the sum of the two products?

- A) 2002 B) 2003 C) 2004 D) 2005.

29) Points P , Q , R and S divide the sides of a rectangle $ABCD$ in the ratio 1 : 2, as shown in the figure. What fraction of area of the rectangle is the area of the parallelogram $PQRS$?



- A) $\frac{2}{5}$ B) $\frac{3}{5}$ C) $\frac{4}{9}$ D) $\frac{5}{9}$

30) One hundred and twenty students take an examination which is marked out of a total 100 (with no fractional marks). No three students are awarded the same mark. The smallest possible *number of pairs* of students who are awarded the same rank is

- A) 9 B) 10 C) 20 D) 19